

Route C: A Spectral Approach to Zero-Density Estimates, Conditional on Two Analytic Gaps

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Disclaimer. This note is *conditional*: two analytic gaps remain open. We do *not* claim a proof of the Riemann Hypothesis.

Abstract

Let $Q_N[i, j] = 1/(\gcd(i, j)\sqrt{ij})$. Numerics suggest $\lambda_{\min}(Q_N)/\log N \rightarrow -1/(2\pi)$ and that the minimizing eigenvector v_N^* stays close to the normalized alternating vector $\hat{v}_{\text{alt}}[k] = (-1)^{k+1}/\sqrt{k}$. We outline a Route C architecture: local Möbius control (Bombieri–Vinogradov scale), a ring/gradient obstruction to sparse concentration, and Davis–Kahan perturbation across a spectral gap. Closing the architecture would yield a polynomial proximity bound $\|v_N^* - \hat{v}_{\text{alt}}\| \ll N^{-\eta}$ for some $\eta > 0$, and thence the spectral-limit conjecture — *conditional* on two gaps stated below. We do not claim those gaps are closed, and we do not claim RH.

1 Operator and numerical targets

$$Q_N[i, j] = \frac{1}{\gcd(i, j)\sqrt{ij}}, \quad 1 \leq i, j \leq N.$$

Write $v_{\text{alt}}[k] = (-1)^{k+1}/\sqrt{k}$ and $\hat{v}_{\text{alt}} = v_{\text{alt}}/\|v_{\text{alt}}\|$. Observed for moderate N (to $N = 200$ in archived checks):

- $\lambda_{\min}(Q_N)/\log N$ tracks $-1/(2\pi)$;
- $|\langle v_N^*, \hat{v}_{\text{alt}} \rangle|$ stays large;
- a positive gap $\lambda_2 - \lambda_{\min}$ appears to persist.

These are numerical observations, not theorems.

2 Architecture (conditional)

Theorem 2.1 (Route C, conditional template). *Assume Gap A (local Möbius / parity evaluation) and Gap B (uniform spectral-gap lower bound). Then there exist $C, \eta > 0$ such that for all large N ,*

$$\|v_N^* - \hat{v}_{\text{alt}}\| \leq C N^{-\eta}.$$

In particular, under the same gaps,

$$\frac{\lambda_{\min}(Q_N)}{\log N} \rightarrow -\frac{1}{2\pi}.$$

Remark 2.2. A standard bridge from a sufficiently strong spectral-floor statement to zero-density / Mertens-type bounds (and thence RH via classical criteria) is logically separate. Even if Gaps A–B close, that bridge must be cited and checked independently. This note does not package RH as proved.

3 Inputs already usable as lemmas

Lemma 3.1 (Local Möbius sketch). *On blocks of length $M = N^{1/4}$, Bombieri–Vinogradov-scale equidistribution suggests a deviation bound of order $M^{5/6}$ after splitting small / medium / large divisors. Making this fully uniform in the Route C ranges is part of Gap A.*

Lemma 3.2 (Concentration cost). *Sparse concentration of a trial vector on “bad” blocks forces large gradient / oscillation cost relative to a spectral gap. In the fluids companion this is the Ring Lemma; here it is only an analogy until Gap B supplies a proved gap.*

Analytic gaps (open). 1. **Gap A.** *Rigorous evaluation $v_{\text{alt}}^\top Q_N v_{\text{alt}} \sim -\log N/(2\pi)$ from the even- d Möbius subseries, without assuming the target constant from zero-density formulae.*

2. **Gap B.** *Analytic lower bound $\lambda_2(Q_N) - \lambda_{\min}(Q_N) \geq c_0 > 0$ uniform in N (currently numeric).*

4 What is not claimed

- RH, or any unconditional zero-density theorem.
- Full-spectrum Bridge floors of the form $\lambda_{\min}(Q_N) > -1/2$ (false for moderate N on related normalizations).
- Closure of Gaps A–B in this draft.

References

- [1] J.E. Littlewood, *Quelques conséquences de l’hypothèse que la fonction $\zeta(s)$ de Riemann n’a pas de zéros dans le demi-plan $\text{Re}(s) > 1/2$* , C. R. Acad. Sci. Paris **154** (1912).
- [2] E. Bombieri, *On the large sieve*, Mathematika **12** (1965), 201–225.
- [3] J.R. Simons, *Route C* (archived draft), Zenodo 10.5281/zenodo.20518388.